King Fahd University of Petroleum and Minerals
Department of Mathematics and Statistics

Math 101
Exam II
Summer
Tuesday 17/07/2012

EXAM COVER

Number of versions: 4
Number of questions: 20
Number of Answers: 5 per question

This exam was prepared using mcqs
For questions send an email to Dr. Ibrahim Al-Lehyani (iallehyani@kaau.edu.sa)

King Fahd University of Petroleum and Minerals
Department of Mathematics and Statistics

Math 101
Exam II
Summer
Tuesday 17/07/2012
Net Time Allowed: 120 minutes

MASTER VERSION

1. If $f(x) = x^e + e^{-2x} + x^{-3x}$, then $f'(x)$ is equal to

(a) $ex^{e-1} - 2e^{-2x} - 3x^{-3x}(1 + \ln x)$

(b) $ex^{e-1} - 2e^{-2x} - 3x^{-3x} \ln x$

(c) $ex^{e-1} - 2e^{-2x} - 3x^{-3x}$

(d) $ex^e + 2e^{-2x} + x^{-3x}(1 + \ln x)$

(e) $ex^{e-1} + 2e^{-2x} - x^{-3x} \ln x$

2. $\lim_{x \to 1} \dfrac{e^{(x^2)} - e}{x - 1} =$ (Hint: this limit is a derivative)

(a) $2e$

(b) e

(c) $-e$

(d) 0

(e) does not exist

3. If $f(x) = \dfrac{\cos x}{1 - \sin x}$, then $f'\left(\dfrac{\pi}{4}\right)$ is equal to

(a) $2 + \sqrt{2}$

(b) $2 - \sqrt{2}$

(c) $\dfrac{1}{\sqrt{2} - 1}$

(d) $\dfrac{1}{\sqrt{2}}$

(e) 2

4. If $f(x) = (\csc 3x + \cot 3x)(\csc 3x - \cot 3x)$, then $f'(x)$ is equal to

(a) 0

(b) $-3 \csc 3x \cot 3x$

(c) $-3 \csc^2 3x$

(d) $-3 \cot^2 3x$

(e) 3

5. The number of points on the curve $y = \dfrac{(x+1)(3x-1)}{x-2}$ where the tangent line is horizontal is

(a) 2

(b) 3

(c) 1

(d) 0

(e) 4

6. If α, β and γ are constants such that the function
$f(x) = \begin{cases} \alpha x^2 + \beta x + \gamma & x < 0 \\ 2\cos x + \tan x & x \geq 0 \end{cases}$
is differentiable everywhere up to the second order, then $f(-1)$ is equal to

(a) 0

(b) -1

(c) -2

(d) 1

(e) 3

7. Find the derivative of $f(x) = \dfrac{\ln(\cos 4x)}{1+x}$ at $x = \dfrac{\pi}{2}$

(a) 0

(b) 2

(c) -4

(d) 1

(e) -1

8. If $y = t^2 - t + 2$ and $x = t^3 + t$, then the value of $\dfrac{dy}{dx}|_{t=1}$ is equal to

(a) $\dfrac{1}{4}$

(b) $\dfrac{1}{2}$

(c) $\dfrac{1}{3}$

(d) $\dfrac{2}{3}$

(e) 2

9. What is the slope of the tangent line to the curve $x + \frac{\sqrt{2}}{4}\cos(xy) = y^2 + \frac{\pi}{2}$ at the point $\left(\frac{\pi}{2}, \frac{1}{2}\right)$

(a) $\frac{7}{8+\pi}$

(b) $\frac{8}{8-\pi}$

(c) $\frac{4(2-\pi)}{8-\pi}$

(d) $\frac{4(2+\pi)}{8-\pi}$

(e) $\frac{7-4\pi}{8+\pi}$

10. Find $\lim_{x\to 0}(1+3x)^{\frac{1}{2x}} =$

(a) $e^{\frac{3}{2}}$

(b) $e^{\frac{2}{3}}$

(c) e^2

(d) e^3

(e) $e^{\frac{1}{2}}$

11. Find the derivative of $y = (\sin x)^{\csc^3 x}$ at $x = \dfrac{\pi}{6}$.
[Hint: Use Logarithmic differentiation]

(a) $\dfrac{\sqrt{3}}{2^5}(1 + 3\ln 2)$

(b) $\dfrac{1}{4}$

(c) $\dfrac{1}{2}(3\ln 2 + 1)$

(d) $\sqrt{3} + \ln 2$

(e) $\dfrac{\sqrt{3}}{2^8}(3 + \ln 2)$

12. If $y = x^2 \ln|x|$ then $y^{(50)}(1) =$

(a) $-2(47!)$

(b) $2(49!)$

(c) $-2(46!)$

(d) $2(45!)$

(e) $2(48!)$

13. The position in meters of a particle moving in a straight line is given by

$$s(t) = 2\sin^2(\pi t/4) - 2\cos^2(\pi t/4)$$

in the time interval $[0, 3]$ seconds. The particle is at rest some time after initial release; at this time let its position be S, and its acceleration be A. Then the product AS is equal to,

(a) $-\pi^2\, m^2/s^2$

(b) $-\pi\, m^2/s^2$

(c) $2\, m^2/s^2$

(d) $\pi\, m^2/s^2$

(e) $\pi^2\, m^2/s^2$

14. A man is standing at the edge of a cliff on the moon. He throws a rock upwards. The vertical position of the rock in meters, from where he is standing, after t seconds is $h(t) = 6t - 0.5t^2\ m$. The rock eventually falls and hits the surface of the moon which is 80 metres below the edge of the cliff. The velocity of the rock when it hits the ground is,

(a) $-14m/s$

(b) $12m/s$

(c) $20m/s$

(d) $-6m/s$

(e) $12m/s$

15. A car A is travelling east at 68 km/h and a car B is travelling south at 80 km/h. Both are headed towards the same intersection of the roads. At what velocity are the cars approaching each other when car A is 500 m and car B is 1200m from the intersection?

(a) -100km/h

(b) -50 km/h

(c) -25 km/h

(d) 35 km/h

(e) 75 km/h

16. A street-light is mounted at the top of a 10-metre-tall pole. A man 2 meters tall runs away from the pole in a straight line with a speed of 8 m/s. At what rate is the line from the top of the pole to the tip of his shadow rotating when he is 6 meters from the pole?

(a) 0.64 rad/s

(b) 0.36 rad/s

(c) 0.25 rad/s

(d) 0.16 rad/s

(e) 1.00 rad/s

17. Given $f(x) = (1 + 2x)^{5.5}e^{-4.4x}$, then using *linearization* of $f(x)$ at $x = 0$, an approximate value of $f(0.1)$ is

(a) $f(0.1) \approx 1.66$

(b) $f(0.1) \approx 1.60$

(c) $f(0.1) \approx 1.10$

(d) $f(0.1) \approx 1.11$

(e) $f(0.1) \approx 2.10$

18. The cone angle θ of a circular cone (see diagram below) is measured and found to be 45^0 with a possible error of at most 0.45^0. If the radius of the circular base r can be measured accurately, what is the maximum percentage error in computing the volume of the cone.
[Hint: the volume of a circular cone of base radius r and height h is $V = \frac{1}{3}\pi r^2 h$.]

(a) $\frac{\pi}{2}\%$

(b) 2%

(c) $\frac{\pi}{4}\%$

(d) $\pi\%$

(e) 1%

19. Find the equation of the normal line to the curve $f(x)$ at $x = 1$ if $f(x) = \sqrt{2}\cos\left(\arctan(x^3)\right)$

(a) $3y = 2x + 1$

(b) $3y = 2x$

(c) $2y = 3x - 2$

(d) $y = 3x - 3$

(e) $2y = x - 1$

20. The position of a particle moving along a straight line is given by $s(t) = \frac{1}{3}t^3 - \frac{3}{2}t^2 + 2t$ in the time interval [0,4] seconds. The time interval when the particle is speeding up is,

(a) $(1, 1.5) \cup (2, 4)$ seconds

(b) $(0, 1) \cup (2, 3)$ seconds

(c) $(0, 1.5) \cup (3, 4)$ seconds

(d) $(1, 2)$ seconds

(e) $(3, 4)$ seconds

King Fahd University of Petroleum and Minerals
Department of Mathematics and Statistics

CODE 001 **Math 101** **CODE 001**

Exam II

Summer

Tuesday 17/07/2012

Net Time Allowed: 120 minutes

Name: _____

ID: _____ Sec: _____.

Check that this exam has 20 questions.

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3. Use a good eraser. DO NOT use the erasers attached to the pencil.
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5. When bubbling your ID number and Section number, be sure that the bubbles match with the numbers that you write.
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7. When bubbling, make sure that the bubbled space is fully covered.
8. When erasing a bubble, make sure that you do not leave any trace of penciling.

1. If $f(x) = (\csc 3x + \cot 3x)(\csc 3x - \cot 3x)$, then $f'(x)$ is equal to

(a) 3

(b) $-3 \csc 3x \cot 3x$

(c) $-3 \csc^2 3x$

(d) $-3 \cot^2 3x$

(e) 0

2. $\lim_{x \to 1} \dfrac{e^{(x^2)} - e}{x - 1} =$ (Hint: this limit is a derivative)

(a) e

(b) $2e$

(c) does not exist

(d) $-e$

(e) 0

3. If $f(x) = \dfrac{\cos x}{1 - \sin x}$, then $f'\left(\dfrac{\pi}{4}\right)$ is equal to

(a) 2

(b) $\dfrac{1}{\sqrt{2}}$

(c) $2 + \sqrt{2}$

(d) $2 - \sqrt{2}$

(e) $\dfrac{1}{\sqrt{2} - 1}$

4. If $f(x) = x^e + e^{-2x} + x^{-3x}$, then $f'(x)$ is equal to

(a) $ex^{e-1} - 2e^{-2x} - 3x^{-3x}(1 + \ln x)$

(b) $ex^{e-1} - 2e^{-2x} - 3x^{-3x}$

(c) $ex^{e-1} - 2e^{-2x} - 3x^{-3x} \ln x$

(d) $ex^{e-1} + 2e^{-2x} - x^{-3x} \ln x$

(e) $ex^e + 2e^{-2x} + x^{-3x}(1 + \ln x)$

5. The number of points on the curve $y = \dfrac{(x+1)(3x-1)}{x-2}$ where the tangent line is horizontal is

(a) 0

(b) 3

(c) 2

(d) 4

(e) 1

6. Find the derivative of $f(x) = \dfrac{\ln(\cos 4x)}{1+x}$ at $x = \dfrac{\pi}{2}$

(a) 2

(b) -1

(c) -4

(d) 1

(e) 0

7. Find $\lim_{x\to 0}(1+3x)^{\frac{1}{2x}} =$

(a) $e^{\frac{1}{2}}$

(b) e^{3}

(c) $e^{\frac{2}{3}}$

(d) e^{2}

(e) $e^{\frac{3}{2}}$

8. If $y = t^2 - t + 2$ and $x = t^3 + t$, then the value of $\frac{dy}{dx}|_{t=1}$ is equal to

(a) $\frac{2}{3}$

(b) $\frac{1}{3}$

(c) 2

(d) $\frac{1}{2}$

(e) $\frac{1}{4}$

9. If α, β and γ are constants such that the function
$f(x) = \begin{cases} \alpha x^2 + \beta x + \gamma & x < 0 \\ 2\cos x + \tan x & x \geq 0 \end{cases}$
is differentiable everywhere up to the second order, then $f(-1)$ is equal to

(a) -2

(b) 1

(c) 3

(d) -1

(e) 0

10. What is the slope of the tangent line to the curve
$x + \frac{\sqrt{2}}{4}\cos(xy) = y^2 + \frac{\pi}{2}$ at the point $\left(\frac{\pi}{2}, \frac{1}{2}\right)$

(a) $\frac{8}{8-\pi}$

(b) $\frac{4(2+\pi)}{8-\pi}$

(c) $\frac{4(2-\pi)}{8-\pi}$

(d) $\frac{7}{8+\pi}$

(e) $\frac{7-4\pi}{8+\pi}$

11. The position in meters of a particle moving in a straight line is given by

$$s(t) = 2\sin^2(\pi t/4) - 2\cos^2(\pi t/4)$$

in the time interval $[0, 3]$ seconds. The particle is at rest some time after initial release; at this time let its position be S, and its acceleration be A. Then the product AS is equal to,

(a) $\pi\, m^2/s^2$

(b) $\pi^2\, m^2/s^2$

(c) $-\pi\, m^2/s^2$

(d) $-\pi^2\, m^2/s^2$

(e) $2\, m^2/s^2$

12. A man is standing at the edge of a cliff on the moon. He throws a rock upwards. The vertical position of the rock in meters, from where he is standing, after t seconds is $h(t) = 6t - 0.5t^2\ m$. The rock eventually falls and hits the surface of the moon which is 80 metres below the edge of the cliff. The velocity of the rock when it hits the ground is,

(a) $20m/s$

(b) $-14m/s$

(c) $12m/s$

(d) $12m/s$

(e) $-6m/s$

13. Find the derivative of $y = (\sin x)^{\csc^3 x}$ at $x = \dfrac{\pi}{6}$.
[Hint: Use Logarithmic differentiation]

(a) $\dfrac{\sqrt{3}}{2^5}(1 + 3\ln 2)$

(b) $\dfrac{1}{4}$

(c) $\dfrac{1}{2}(3\ln 2 + 1)$

(d) $\sqrt{3} + \ln 2$

(e) $\dfrac{\sqrt{3}}{2^8}(3 + \ln 2)$

14. If $y = x^2 \ln|x|$ then $y^{(50)}(1) =$

(a) $2(45!)$

(b) $2(48!)$

(c) $-2(47!)$

(d) $-2(46!)$

(e) $2(49!)$

15. A car A is travelling east at 68 km/h and a car B is travelling south at 80 km/h. Both are headed towards the same intersection of the roads. At what velocity are the cars approaching each other when car A is 500 m and car B is 1200m from the intersection?

(a) 35 km/h

(b) -100km/h

(c) -50 km/h

(d) -25 km/h

(e) 75 km/h

16. A street-light is mounted at the top of a 10-metre-tall pole. A man 2 meters tall runs away from the pole in a straight line with a speed of 8 m/s. At what rate is the line from the top of the pole to the tip of his shadow rotating when he is 6 meters from the pole?

(a) 0.36 rad/s

(b) 1.00 rad/s

(c) 0.16 rad/s

(d) 0.25 rad/s

(e) 0.64 rad/s

17. Given $f(x) = (1 + 2x)^{5.5}e^{-4.4x}$, then using *linearization* of $f(x)$ at $x = 0$, an approximate value of $f(0.1)$ is

(a) $f(0.1) \approx 1.66$

(b) $f(0.1) \approx 1.11$

(c) $f(0.1) \approx 1.10$

(d) $f(0.1) \approx 2.10$

(e) $f(0.1) \approx 1.60$

18. The cone angle θ of a circular cone (see diagram below) is measured and found to be 45^0 with a possible error of at most 0.45^0. If the radius of the circular base r can be measured accurately, what is the maximum percentage error in computing the volume of the cone.
[Hint: the volume of a circular cone of base radius r and height h is $V = \frac{1}{3}\pi r^2 h$.]

(a) $\frac{\pi}{2}\%$

(b) $\frac{\pi}{4}\%$

(c) $\pi\%$

(d) 1%

(e) 2%

19. The position of a particle moving along a straight line is given by $s(t) = \frac{1}{3}t^3 - \frac{3}{2}t^2 + 2t$ in the time interval [0,4] seconds. The time interval when the particle is speeding up is,

(a) $(1, 1.5) \cup (2, 4)$ seconds

(b) $(1, 2)$ seconds

(c) $(0, 1) \cup (2, 3)$ seconds

(d) $(3, 4)$ seconds

(e) $(0, 1.5) \cup (3, 4)$ seconds

20. Find the equation of the normal line to the curve $f(x)$ at $x = 1$ if $f(x) = \sqrt{2}\cos\left(\arctan(x^3)\right)$

(a) $3y = 2x + 1$

(b) $2y = x - 1$

(c) $y = 3x - 3$

(d) $2y = 3x - 2$

(e) $3y = 2x$

Name .....

ID ..... Sec .....

1	a	b	c	d	e	f
2	a	b	c	d	e	f
3	a	b	c	d	e	f
4	a	b	c	d	e	f
5	a	b	c	d	e	f
6	a	b	c	d	e	f
7	a	b	c	d	e	f
8	a	b	c	d	e	f
9	a	b	c	d	e	f
10	a	b	c	d	e	f
11	a	b	c	d	e	f
12	a	b	c	d	e	f
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14	a	b	c	d	e	f
15	a	b	c	d	e	f
16	a	b	c	d	e	f
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20	a	b	c	d	e	f
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49	a	b	c	d	e	f
50	a	b	c	d	e	f
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53	a	b	c	d	e	f
54	a	b	c	d	e	f
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61	a	b	c	d	e	f
62	a	b	c	d	e	f
63	a	b	c	d	e	f
64	a	b	c	d	e	f
65	a	b	c	d	e	f
66	a	b	c	d	e	f
67	a	b	c	d	e	f
68	a	b	c	d	e	f
69	a	b	c	d	e	f
70	a	b	c	d	e	f

King Fahd University of Petroleum and Minerals
Department of Mathematics and Statistics

CODE 002 **Math 101** **CODE 002**

Exam II
Summer
Tuesday 17/07/2012
Net Time Allowed: 120 minutes

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7. When bubbling, make sure that the bubbled space is fully covered.
8. When erasing a bubble, make sure that you do not leave any trace of penciling.

1. If $f(x) = \dfrac{\cos x}{1 - \sin x}$, then $f'\left(\dfrac{\pi}{4}\right)$ is equal to

(a) 2

(b) $\dfrac{1}{\sqrt{2}}$

(c) $\dfrac{1}{\sqrt{2} - 1}$

(d) $2 - \sqrt{2}$

(e) $2 + \sqrt{2}$

2. The number of points on the curve $y = \dfrac{(x+1)(3x-1)}{x-2}$ where the tangent line is horizontal is

(a) 3

(b) 0

(c) 4

(d) 1

(e) 2

3. If $f(x) = x^e + e^{-2x} + x^{-3x}$, then $f'(x)$ is equal to

(a) $ex^{e-1} - 2e^{-2x} - 3x^{-3x}(1 + \ln x)$

(b) $ex^e + 2e^{-2x} + x^{-3x}(1 + \ln x)$

(c) $ex^{e-1} - 2e^{-2x} - 3x^{-3x} \ln x$

(d) $ex^{e-1} + 2e^{-2x} - x^{-3x} \ln x$

(e) $ex^{e-1} - 2e^{-2x} - 3x^{-3x}$

4. $\lim_{x \to 1} \dfrac{e^{(x^2)} - e}{x - 1} =$ (Hint: this limit is a derivative)

(a) 0

(b) e

(c) does not exist

(d) $-e$

(e) $2e$

5. If $f(x) = (\csc 3x + \cot 3x)(\csc 3x - \cot 3x)$, then $f'(x)$ is equal to

(a) $-3\cot^2 3x$

(b) 3

(c) 0

(d) $-3\csc 3x \cot 3x$

(e) $-3\csc^2 3x$

6. What is the slope of the tangent line to the curve $x + \frac{\sqrt{2}}{4}\cos(xy) = y^2 + \frac{\pi}{2}$ at the point $\left(\frac{\pi}{2}, \frac{1}{2}\right)$

(a) $\frac{7}{8+\pi}$

(b) $\frac{4(2+\pi)}{8-\pi}$

(c) $\frac{4(2-\pi)}{8-\pi}$

(d) $\frac{8}{8-\pi}$

(e) $\frac{7-4\pi}{8+\pi}$

7. Find $\lim_{x\to 0}(1+3x)^{\frac{1}{2x}} =$

(a) e^2

(b) e^3

(c) $e^{\frac{1}{2}}$

(d) $e^{\frac{2}{3}}$

(e) $e^{\frac{3}{2}}$

8. Find the derivative of $f(x) = \dfrac{\ln(\cos 4x)}{1+x}$ at $x = \dfrac{\pi}{2}$

(a) -4

(b) 2

(c) -1

(d) 0

(e) 1

9. If α, β and γ are constants such that the function
$f(x) = \begin{cases} \alpha x^2 + \beta x + \gamma & x < 0 \\ 2\cos x + \tan x & x \geq 0 \end{cases}$
is differentiable everywhere up to the second order, then $f(-1)$ is equal to

(a) 0

(b) -2

(c) -1

(d) 1

(e) 3

10. If $y = t^2 - t + 2$ and $x = t^3 + t$, then the value of $\frac{dy}{dx}|_{t=1}$ is equal to

(a) $\frac{1}{3}$

(b) $\frac{1}{2}$

(c) 2

(d) $\frac{1}{4}$

(e) $\frac{2}{3}$

11. A man is standing at the edge of a cliff on the moon. He throws a rock upwards. The vertical position of the rock in meters, from where he is standing, after t seconds is $h(t) = 6t - 0.5t^2$ m. The rock eventually falls and hits the surface of the moon which is 80 metres below the edge of the cliff. The velocity of the rock when it hits the ground is,

(a) $12m/s$

(b) $-6m/s$

(c) $12m/s$

(d) $20m/s$

(e) $-14m/s$

12. A car A is travelling east at 68 km/h and a car B is travelling south at 80 km/h. Both are headed towards the same intersection of the roads. At what velocity are the cars approaching each other when car A is 500 m and car B is 1200m from the intersection?

(a) 75 km/h

(b) -50 km/h

(c) 35 km/h

(d) -25 km/h

(e) -100km/h

13. The position in meters of a particle moving in a straight line is given by

$$s(t) = 2\sin^2(\pi t/4) - 2\cos^2(\pi t/4)$$

in the time interval $[0, 3]$ seconds. The particle is at rest some time after initial release; at this time let its position be S, and its acceleration be A. Then the product AS is equal to,

(a) $\pi\, m^2/s^2$

(b) $\pi^2\, m^2/s^2$

(c) $2\, m^2/s^2$

(d) $-\pi\, m^2/s^2$

(e) $-\pi^2\, m^2/s^2$

14. Find the derivative of $y = (\sin x)^{\csc^3 x}$ at $x = \dfrac{\pi}{6}$.
[Hint: Use Logarithmic differentiation]

(a) $\dfrac{1}{2}(3\ln 2 + 1)$

(b) $\dfrac{1}{4}$

(c) $\dfrac{\sqrt{3}}{2^5}(1 + 3\ln 2)$

(d) $\sqrt{3} + \ln 2$

(e) $\dfrac{\sqrt{3}}{2^8}(3 + \ln 2)$

15. If $y = x^2 \ln|x|$ then $y^{(50)}(1) =$

(a) $-2(47!)$

(b) $-2(46!)$

(c) $2(49!)$

(d) $2(48!)$

(e) $2(45!)$

16. A street-light is mounted at the top of a 10-metre-tall pole. A man 2 meters tall runs away from the pole in a straight line with a speed of 8 m/s. At what rate is the line from the top of the pole to the tip of his shadow rotating when he is 6 meters from the pole?

(a) 0.36 rad/s

(b) 0.64 rad/s

(c) 1.00 rad/s

(d) 0.25 rad/s

(e) 0.16 rad/s

17. The position of a particle moving along a straight line is given by $s(t) = \frac{1}{3}t^3 - \frac{3}{2}t^2 + 2t$ in the time interval [0,4] seconds. The time interval when the particle is speeding up is,

(a) $(0, 1) \cup (2, 3)$ seconds

(b) $(1, 2)$ seconds

(c) $(3, 4)$ seconds

(d) $(0, 1.5) \cup (3, 4)$ seconds

(e) $(1, 1.5) \cup (2, 4)$ seconds

18. The cone angle θ of a circular cone (see diagram below) is measured and found to be 45^0 with a possible error of at most 0.45^0. If the radius of the circular base r can be measured accurately, what is the maximum percentage error in computing the volume of the cone.
[Hint: the volume of a circular cone of base radius r and height h is $V = \frac{1}{3}\pi r^2 h$.]

(a) 2%

(b) $\frac{\pi}{4}\%$

(c) $\frac{\pi}{2}\%$

(d) 1%

(e) $\pi\%$

19. Find the equation of the normal line to the curve $f(x)$ at $x = 1$ if $f(x) = \sqrt{2}\cos\left(\arctan(x^3)\right)$

(a) $y = 3x - 3$

(b) $2y = x - 1$

(c) $3y = 2x$

(d) $3y = 2x + 1$

(e) $2y = 3x - 2$

20. Given $f(x) = (1 + 2x)^{5.5}e^{-4.4x}$, then using *linearization* of $f(x)$ at $x = 0$, an approximate value of $f(0.1)$ is

(a) $f(0.1) \approx 1.60$

(b) $f(0.1) \approx 1.11$

(c) $f(0.1) \approx 1.66$

(d) $f(0.1) \approx 1.10$

(e) $f(0.1) \approx 2.10$

Name .....

ID ..... Sec .....

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63	a	b	c	d	e	f
64	a	b	c	d	e	f
65	a	b	c	d	e	f
66	a	b	c	d	e	f
67	a	b	c	d	e	f
68	a	b	c	d	e	f
69	a	b	c	d	e	f
70	a	b	c	d	e	f

King Fahd University of Petroleum and Minerals
Department of Mathematics and Statistics

CODE 003 **Math 101** **CODE 003**

Exam II

Summer

Tuesday 17/07/2012

Net Time Allowed: 120 minutes

Name: _____

ID: _____ Sec: _____.

Check that this exam has 20 questions.

Important Instructions:

1. All types of calculators, pagers or mobile phones are NOT allowed during the examination.
2. Use HB 2.5 pencils only.
3. Use a good eraser. DO NOT use the erasers attached to the pencil.
4. Write your name, ID number and Section number on the examination paper and in the upper left corner of the answer sheet.
5. When bubbling your ID number and Section number, be sure that the bubbles match with the numbers that you write.
6. The Test Code Number is already bubbled in your answer sheet. Make sure that it is the same as that printed on your question paper.
7. When bubbling, make sure that the bubbled space is fully covered.
8. When erasing a bubble, make sure that you do not leave any trace of penciling.

1. The number of points on the curve $y = \dfrac{(x+1)(3x-1)}{x-2}$ where the tangent line is horizontal is

(a) 4

(b) 3

(c) 0

(d) 2

(e) 1

2. If $f(x) = \dfrac{\cos x}{1-\sin x}$, then $f'\left(\dfrac{\pi}{4}\right)$ is equal to

(a) $2+\sqrt{2}$

(b) $\dfrac{1}{\sqrt{2}-1}$

(c) $\dfrac{1}{\sqrt{2}}$

(d) $2-\sqrt{2}$

(e) 2

3. If $f(x) = (\csc 3x + \cot 3x)(\csc 3x - \cot 3x)$, then $f'(x)$ is equal to

(a) 0

(b) $-3\csc^2 3x$

(c) 3

(d) $-3\csc 3x \cot 3x$

(e) $-3\cot^2 3x$

4. If $f(x) = x^e + e^{-2x} + x^{-3x}$, then $f'(x)$ is equal to

(a) $ex^{e-1} - 2e^{-2x} - 3x^{-3x}$

(b) $ex^{e-1} - 2e^{-2x} - 3x^{-3x}\ln x$

(c) $ex^e + 2e^{-2x} + x^{-3x}(1 + \ln x)$

(d) $ex^{e-1} - 2e^{-2x} - 3x^{-3x}(1 + \ln x)$

(e) $ex^{e-1} + 2e^{-2x} - x^{-3x}\ln x$

5. $\lim_{x\to 1} \dfrac{e^{(x^2)} - e}{x - 1} =$ (Hint: this limit is a derivative)

(a) 0

(b) e

(c) does not exist

(d) $2e$

(e) $-e$

6. Find $\lim_{x\to 0}(1 + 3x)^{\frac{1}{2x}} =$

(a) $e^{\frac{1}{2}}$

(b) e^3

(c) e^2

(d) $e^{\frac{2}{3}}$

(e) $e^{\frac{3}{2}}$

7. What is the slope of the tangent line to the curve $x + \frac{\sqrt{2}}{4}\cos(xy) = y^2 + \frac{\pi}{2}$ at the point $\left(\frac{\pi}{2}, \frac{1}{2}\right)$

(a) $\frac{8}{8-\pi}$

(b) $\frac{4(2+\pi)}{8-\pi}$

(c) $\frac{7-4\pi}{8+\pi}$

(d) $\frac{7}{8+\pi}$

(e) $\frac{4(2-\pi)}{8-\pi}$

8. Find the derivative of $f(x) = \frac{\ln(\cos 4x)}{1+x}$ at $x = \frac{\pi}{2}$

(a) -1

(b) 0

(c) -4

(d) 1

(e) 2

9. If α, β and γ are constants such that the function
$f(x) = \begin{cases} \alpha x^2 + \beta x + \gamma & x < 0 \\ 2\cos x + \tan x & x \geq 0 \end{cases}$
is differentiable everywhere up to the second order, then $f(-1)$ is equal to

(a) 1

(b) -1

(c) 3

(d) -2

(e) 0

10. If $y = t^2 - t + 2$ and $x = t^3 + t$, then the value of $\frac{dy}{dx}|_{t=1}$ is equal to

(a) $\frac{2}{3}$

(b) 2

(c) $\frac{1}{4}$

(d) $\frac{1}{3}$

(e) $\frac{1}{2}$

11. A car A is travelling east at 68 km/h and a car B is travelling south at 80 km/h. Both are headed towards the same intersection of the roads. At what velocity are the cars approaching each other when car A is 500 m and car B is 1200m from the intersection?

(a) -100km/h

(b) -50 km/h

(c) -25 km/h

(d) 75 km/h

(e) 35 km/h

12. Find the derivative of $y = (\sin x)^{\csc^3 x}$ at $x = \dfrac{\pi}{6}$.
[Hint: Use Logarithmic differentiation]

(a) $\dfrac{1}{2}(3\ln 2 + 1)$

(b) $\dfrac{1}{4}$

(c) $\dfrac{\sqrt{3}}{2^5}(1 + 3\ln 2)$

(d) $\sqrt{3} + \ln 2$

(e) $\dfrac{\sqrt{3}}{2^8}(3 + \ln 2)$

13. A man is standing at the edge of a cliff on the moon. He throws a rock upwards. The vertical position of the rock in meters, from where he is standing, after t seconds is $h(t) = 6t - 0.5t^2$ m. The rock eventually falls and hits the surface of the moon which is 80 metres below the edge of the cliff. The velocity of the rock when it hits the ground is,

(a) $20m/s$

(b) $-14m/s$

(c) $-6m/s$

(d) $12m/s$

(e) $12m/s$

14. The position in meters of a particle moving in a straight line is given by

$$s(t) = 2\sin^2(\pi t/4) - 2\cos^2(\pi t/4)$$

in the time interval $[0, 3]$ seconds. The particle is at rest some time after initial release; at this time let its position be S, and its acceleration be A. Then the product AS is equal to,

(a) $\pi\, m^2/s^2$

(b) $-\pi\, m^2/s^2$

(c) $-\pi^2\, m^2/s^2$

(d) $2\, m^2/s^2$

(e) $\pi^2\, m^2/s^2$

15. If $y = x^2 \ln|x|$ then $y^{(50)}(1) =$

(a) $2(48!)$

(b) $-2(47!)$

(c) $2(49!)$

(d) $-2(46!)$

(e) $2(45!)$

16. Given $f(x) = (1+2x)^{5.5}e^{-4.4x}$, then using *linearization* of $f(x)$ at $x = 0$, an approximate value of $f(0.1)$ is

(a) $f(0.1) \approx 1.10$

(b) $f(0.1) \approx 1.11$

(c) $f(0.1) \approx 1.66$

(d) $f(0.1) \approx 2.10$

(e) $f(0.1) \approx 1.60$

17. The position of a particle moving along a straight line is given by $s(t) = \frac{1}{3}t^3 - \frac{3}{2}t^2 + 2t$ in the time interval [0,4] seconds. The time interval when the particle is speeding up is,

(a) $(1, 2)$ seconds

(b) $(3, 4)$ seconds

(c) $(0, 1.5) \cup (3, 4)$ seconds

(d) $(1, 1.5) \cup (2, 4)$ seconds

(e) $(0, 1) \cup (2, 3)$ seconds

18. A street-light is mounted at the top of a 10-metre-tall pole. A man 2 meters tall runs away from the pole in a straight line with a speed of 8 m/s. At what rate is the line from the top of the pole to the tip of his shadow rotating when he is 6 meters from the pole?

(a) 1.00 rad/s

(b) 0.25 rad/s

(c) 0.16 rad/s

(d) 0.36 rad/s

(e) 0.64 rad/s

19. Find the equation of the normal line to the curve $f(x)$ at $x = 1$ if $f(x) = \sqrt{2}\cos\left(\arctan(x^3)\right)$

(a) $3y = 2x + 1$

(b) $2y = x - 1$

(c) $2y = 3x - 2$

(d) $y = 3x - 3$

(e) $3y = 2x$

20. The cone angle θ of a circular cone (see diagram below) is measured and found to be 45^0 with a possible error of at most 0.45^0. If the radius of the circular base r can be measured accurately, what is the maximum percentage error in computing the volume of the cone.
[Hint: the volume of a circular cone of base radius r and height h is $V = \frac{1}{3}\pi r^2 h$.]

(a) $\frac{\pi}{4}\%$

(b) 1%

(c) $\pi\%$

(d) 2%

(e) $\frac{\pi}{2}\%$

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67	a	b	c	d	e	f
68	a	b	c	d	e	f
69	a	b	c	d	e	f
70	a	b	c	d	e	f

King Fahd University of Petroleum and Minerals
Department of Mathematics and Statistics

CODE 004 **Math 101** **CODE 004**

Exam II

Summer

Tuesday 17/07/2012

Net Time Allowed: 120 minutes

Name: _____

ID: _____ Sec: _____.

Check that this exam has 20 questions.

Important Instructions:

1. All types of calculators, pagers or mobile phones are NOT allowed during the examination.
2. Use HB 2.5 pencils only.
3. Use a good eraser. DO NOT use the erasers attached to the pencil.
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7. When bubbling, make sure that the bubbled space is fully covered.
8. When erasing a bubble, make sure that you do not leave any trace of penciling.

1. If $f(x) = x^e + e^{-2x} + x^{-3x}$, then $f'(x)$ is equal to

(a) $ex^{e-1} + 2e^{-2x} - x^{-3x} \ln x$

(b) $ex^{e-1} - 2e^{-2x} - 3x^{-3x}(1 + \ln x)$

(c) $ex^{e-1} - 2e^{-2x} - 3x^{-3x}$

(d) $ex^{e-1} - 2e^{-2x} - 3x^{-3x} \ln x$

(e) $ex^e + 2e^{-2x} + x^{-3x}(1 + \ln x)$

2. If $f(x) = \dfrac{\cos x}{1 - \sin x}$, then $f'\left(\dfrac{\pi}{4}\right)$ is equal to

(a) $2 - \sqrt{2}$

(b) $\dfrac{1}{\sqrt{2} - 1}$

(c) $\dfrac{1}{\sqrt{2}}$

(d) 2

(e) $2 + \sqrt{2}$

3. $\lim_{x\to 1} \dfrac{e^{(x^2)} - e}{x - 1} =$ (Hint: this limit is a derivative)

(a) 0

(b) e

(c) $2e$

(d) does not exist

(e) $-e$

4. If $f(x) = (\csc 3x + \cot 3x)(\csc 3x - \cot 3x)$, then $f'(x)$ is equal to

(a) 0

(b) $-3\csc^2 3x$

(c) 3

(d) $-3\cot^2 3x$

(e) $-3\csc 3x \cot 3x$

5. The number of points on the curve $y = \dfrac{(x+1)(3x-1)}{x-2}$ where the tangent line is horizontal is

(a) 1

(b) 0

(c) 2

(d) 4

(e) 3

6. What is the slope of the tangent line to the curve $x + \dfrac{\sqrt{2}}{4}\cos(xy) = y^2 + \dfrac{\pi}{2}$ at the point $\left(\dfrac{\pi}{2}, \dfrac{1}{2}\right)$

(a) $\dfrac{4(2+\pi)}{8-\pi}$

(b) $\dfrac{7-4\pi}{8+\pi}$

(c) $\dfrac{4(2-\pi)}{8-\pi}$

(d) $\dfrac{7}{8+\pi}$

(e) $\dfrac{8}{8-\pi}$

7. If α, β and γ are constants such that the function
$f(x) = \begin{cases} \alpha x^2 + \beta x + \gamma & x < 0 \\ 2\cos x + \tan x & x \geq 0 \end{cases}$
is differentiable everywhere up to the second order, then $f(-1)$ is equal to

(a) 0

(b) -1

(c) 3

(d) 1

(e) -2

8. If $y = t^2 - t + 2$ and $x = t^3 + t$, then the value of $\frac{dy}{dx}|_{t=1}$ is equal to

(a) $\frac{1}{2}$

(b) $\frac{1}{4}$

(c) $\frac{2}{3}$

(d) 2

(e) $\frac{1}{3}$

9. Find the derivative of $f(x) = \dfrac{\ln(\cos 4x)}{1+x}$ at $x = \dfrac{\pi}{2}$

(a) 2

(b) -4

(c) -1

(d) 1

(e) 0

10. Find $\lim\limits_{x\to 0}(1+3x)^{\frac{1}{2x}} =$

(a) e^3

(b) $e^{\frac{3}{2}}$

(c) e^2

(d) $e^{\frac{2}{3}}$

(e) $e^{\frac{1}{2}}$

11. A car A is travelling east at 68 km/h and a car B is travelling south at 80 km/h. Both are headed towards the same intersection of the roads. At what velocity are the cars approaching each other when car A is 500 m and car B is 1200m from the intersection?

(a) 75 km/h

(b) -100km/h

(c) 35 km/h

(d) -25 km/h

(e) -50 km/h

12. A man is standing at the edge of a cliff on the moon. He throws a rock upwards. The vertical position of the rock in meters, from where he is standing, after t seconds is $h(t) = 6t - 0.5t^2$ m. The rock eventually falls and hits the surface of the moon which is 80 metres below the edge of the cliff. The velocity of the rock when it hits the ground is,

(a) $-14m/s$

(b) $20m/s$

(c) $-6m/s$

(d) $12m/s$

(e) $12m/s$

13. The position in meters of a particle moving in a straight line is given by

$$s(t) = 2\sin^2(\pi t/4) - 2\cos^2(\pi t/4)$$

in the time interval $[0, 3]$ seconds. The particle is at rest some time after initial release; at this time let its position be S, and its acceleration be A. Then the product AS is equal to,

(a) $-\pi^2\, m^2/s^2$

(b) $2\, m^2/s^2$

(c) $\pi\, m^2/s^2$

(d) $\pi^2\, m^2/s^2$

(e) $-\pi\, m^2/s^2$

14. Find the derivative of $y = (\sin x)^{\csc^3 x}$ at $x = \dfrac{\pi}{6}$.
[Hint: Use Logarithmic differentiation]

(a) $\dfrac{\sqrt{3}}{2^5}(1 + 3\ln 2)$

(b) $\dfrac{1}{4}$

(c) $\dfrac{\sqrt{3}}{2^8}(3 + \ln 2)$

(d) $\sqrt{3} + \ln 2$

(e) $\dfrac{1}{2}(3\ln 2 + 1)$

15. If $y = x^2 \ln|x|$ then $y^{(50)}(1) =$

(a) $-2(46!)$

(b) $2(48!)$

(c) $2(45!)$

(d) $-2(47!)$

(e) $2(49!)$

16. The position of a particle moving along a straight line is given by $s(t) = \frac{1}{3}t^3 - \frac{3}{2}t^2 + 2t$ in the time interval [0,4] seconds. The time interval when the particle is speeding up is,

(a) $(0, 1.5) \cup (3, 4)$ seconds

(b) $(3, 4)$ seconds

(c) $(1, 1.5) \cup (2, 4)$ seconds

(d) $(0, 1) \cup (2, 3)$ seconds

(e) $(1, 2)$ seconds

17. The cone angle θ of a circular cone (see diagram below) is measured and found to be 45^0 with a possible error of at most 0.45^0. If the radius of the circular base r can be measured accurately, what is the maximum percentage error in computing the volume of the cone.
[Hint: the volume of a circular cone of base radius r and height h is $V = \frac{1}{3}\pi r^2 h$.]

(a) $\pi\%$

(b) 2%

(c) $\frac{\pi}{2}\%$

(d) 1%

(e) $\frac{\pi}{4}\%$

18. Find the equation of the normal line to the curve $f(x)$ at $x = 1$ if $f(x) = \sqrt{2}\cos\big(\arctan(x^3)\big)$

(a) $y = 3x - 3$

(b) $2y = x - 1$

(c) $3y = 2x$

(d) $2y = 3x - 2$

(e) $3y = 2x + 1$

19. Given $f(x) = (1 + 2x)^{5.5}e^{-4.4x}$, then using *linearization* of $f(x)$ at $x = 0$, an approximate value of $f(0.1)$ is

(a) $f(0.1) \approx 1.11$

(b) $f(0.1) \approx 1.66$

(c) $f(0.1) \approx 1.60$

(d) $f(0.1) \approx 1.10$

(e) $f(0.1) \approx 2.10$

20. A street-light is mounted at the top of a 10-metre-tall pole. A man 2 meters tall runs away from the pole in a straight line with a speed of 8 m/s. At what rate is the line from the top of the pole to the tip of his shadow rotating when he is 6 meters from the pole?

(a) 0.16 rad/s

(b) 0.64 rad/s

(c) 0.25 rad/s

(d) 1.00 rad/s

(e) 0.36 rad/s

Name .....

ID ..... Sec .....

1	a	b	c	d	e	f
2	a	b	c	d	e	f
3	a	b	c	d	e	f
4	a	b	c	d	e	f
5	a	b	c	d	e	f
6	a	b	c	d	e	f
7	a	b	c	d	e	f
8	a	b	c	d	e	f
9	a	b	c	d	e	f
10	a	b	c	d	e	f
11	a	b	c	d	e	f
12	a	b	c	d	e	f
13	a	b	c	d	e	f
14	a	b	c	d	e	f
15	a	b	c	d	e	f
16	a	b	c	d	e	f
17	a	b	c	d	e	f
18	a	b	c	d	e	f
19	a	b	c	d	e	f
20	a	b	c	d	e	f
21	a	b	c	d	e	f
22	a	b	c	d	e	f
23	a	b	c	d	e	f
24	a	b	c	d	e	f
25	a	b	c	d	e	f
26	a	b	c	d	e	f
27	a	b	c	d	e	f
28	a	b	c	d	e	f
29	a	b	c	d	e	f
30	a	b	c	d	e	f
31	a	b	c	d	e	f
32	a	b	c	d	e	f
33	a	b	c	d	e	f
34	a	b	c	d	e	f
35	a	b	c	d	e	f

36	a	b	c	d	e	f
37	a	b	c	d	e	f
38	a	b	c	d	e	f
39	a	b	c	d	e	f
40	a	b	c	d	e	f
41	a	b	c	d	e	f
42	a	b	c	d	e	f
43	a	b	c	d	e	f
44	a	b	c	d	e	f
45	a	b	c	d	e	f
46	a	b	c	d	e	f
47	a	b	c	d	e	f
48	a	b	c	d	e	f
49	a	b	c	d	e	f
50	a	b	c	d	e	f
51	a	b	c	d	e	f
52	a	b	c	d	e	f
53	a	b	c	d	e	f
54	a	b	c	d	e	f
55	a	b	c	d	e	f
56	a	b	c	d	e	f
57	a	b	c	d	e	f
58	a	b	c	d	e	f
59	a	b	c	d	e	f
60	a	b	c	d	e	f
61	a	b	c	d	e	f
62	a	b	c	d	e	f
63	a	b	c	d	e	f
64	a	b	c	d	e	f
65	a	b	c	d	e	f
66	a	b	c	d	e	f
67	a	b	c	d	e	f
68	a	b	c	d	e	f
69	a	b	c	d	e	f
70	a	b	c	d	e	f

Q	MM	V1	V2	V3	V4
1	a	e	e	d	b
2	a	b	e	a	e
3	a	c	a	a	c
4	a	a	e	d	a
5	a	c	c	d	c
6	a	e	a	e	d
7	a	e	e	d	a
8	a	e	d	b	b
9	a	e	a	e	e
10	a	d	d	c	b
11	a	d	e	a	b
12	a	b	e	c	a
13	a	a	e	b	a
14	a	c	c	c	a
15	a	b	a	b	d
16	a	e	b	c	c
17	a	a	e	d	c
18	a	a	c	e	e
19	a	a	d	a	b
20	a	a	c	e	b

Answer Counts

V	a	b	c	d	e
1	3	3	3	6	5
2	1	5	3	8	3
3	3	7	3	3	4
4	6	4	5	1	4